

Module-4

ASSOCIATED IOT TECHNOLOGIES

Cloud Computing:
Introduction,
Virtualization,
Cloud Models,
Service-Level Agreement in Cloud Computing,
Cloud Implementation,
Sensor-Cloud:
Sensors-as-a-Service.

IOT CASE STUDIES

Agricultural IoT – Introduction and Case Studies

Textbook 1: Chapter 10– 10.1 to 10.6; Chapter 12- 12.1-12.2

4.1 Introduction

- Sensor nodes are the key components of Internet of Things (IoT).
- These nodes are resource-constrained in terms of storage, processing, and energy.
- The devices are connected and communicate with one another by sharing the sensed and processed data.
- Handling the enormous data generated by this large number of heterogeneous devices is a non-trivial task.
- Consequently, cloud computing becomes an essential building block of the IoT architecture.
- Cloud computing is more than traditional network computing.
- Cloud computing comprises a pool of multiple resources such as servers, storage, and network from single/multiple organizations.
- These resources are allocated to the end users as per requirement, on a payment basis.
- In cloud computing architecture, an end user can request for customized resources such as storage space, RAM, operating systems, and other software to a cloud service provider (CSP) as shown in Figure 10.1.
- For example, a user can request for a Linux operating system for running an application from a CSP; another end user can request for Windows 10 operating system from the same CSP for executing some application.
- The cloud services are accessible from anywhere and at any time by an authorized user through Internet connectivity.

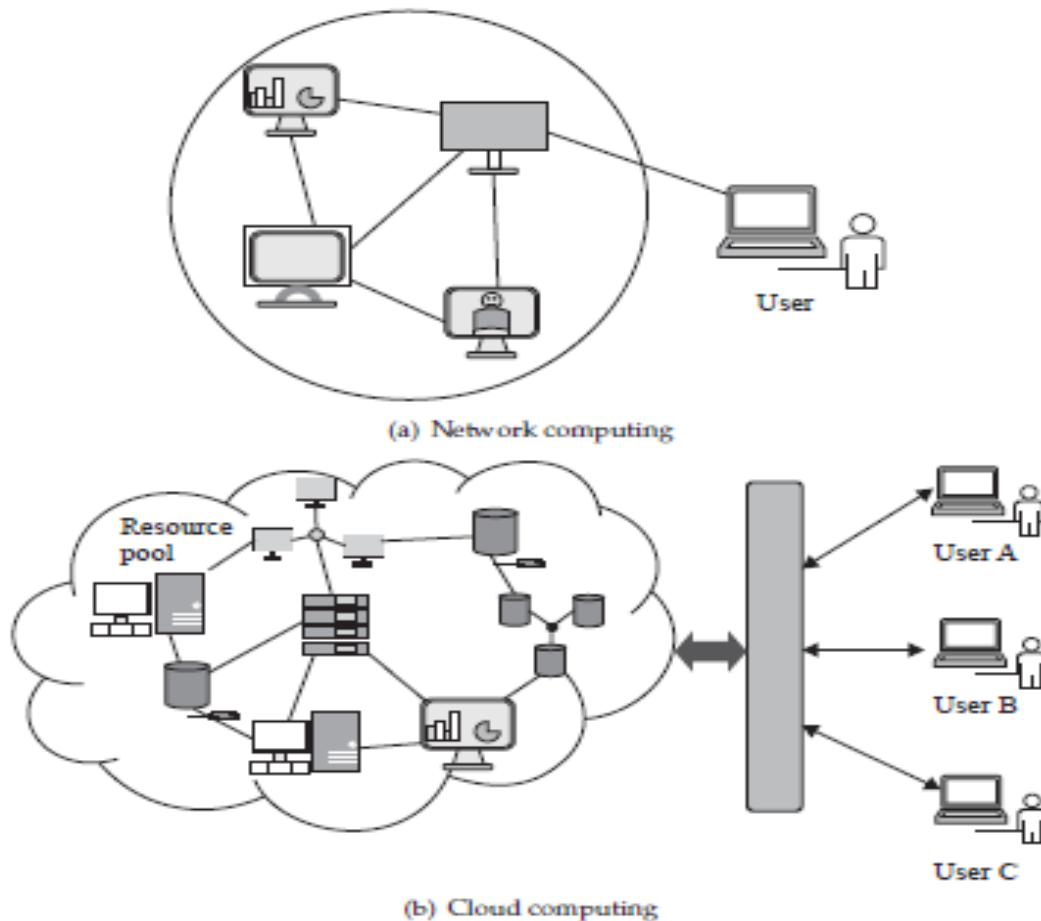


Figure 10.1 Network computing versus cloud computing

- Cloud computing comprises a shared pool of computing resources, which are accessible dynamically, ubiquitously, and on-demand basis by the users.
- This shared pool of resources includes networks, storage, processor, and servers.
- These resources are accessible by multiple users through a regular command-line terminal at the same or different time instants.
- The services of cloud computing are based on the pay-per-use model.
- The concept is the same as paying utility bills based on consumption.
- In cloud computing, a user pays for the cloud services as per the duration of their resource usage.
- On the other hand, there is a CSP, that provides cloud services to end user organizations.

4.2 Virtualization

- The key concept of cloud computing is virtualization.
- The technique of sharing a single resource among multiple end user organizations or end users is known as virtualization.
- In the virtualization process, a physical resource is logically distributed among multiple users.
- However, a user perceives that the resource is unlimited and is dedicatedly provided to him/her.
- Figure 10.2(a) represents a traditional desktop, where an application (App) is running on top of an OS, and resources are utilized only for that particular application.
- On the other hand, multiple resources can be used by different end users through virtualization software, as shown in Figure 10.2(b).
- Virtualization software separates the resources logically so that there is no conflict among the users during resource utilization.

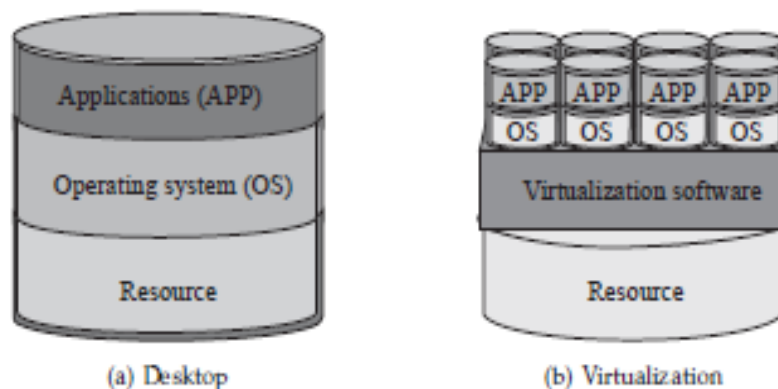


Figure 10.2 Traditional desktop versus virtualization

4.2.1 Advantages of virtualization

- With the increasing number of interconnected heterogeneous devices in IoT, the importance of virtualization also increases.
- There are different software such as VMware, which enable the concept of virtualization.

- With the increasing importance of cloud computing, different organizations and individuals are using it extensively.
- There is always a risk of system crash at any instant of time.
- In such a scenario, cloud computing plays a vital role by keeping backups through virtualization.
- Primarily, there are two entities in a cloud computing architecture:
 1. End users and
 2. CSP.
- Both end users and CSP are benefited in several aspects through the process of virtualization.
- The major advantages, from the perspective of the end user and CSP, are as follows:

(i) Advantages for End Users

(a) Variety:

- The process of virtualization in cloud computing enables an end user organization to use various types of applications based on the requirements.
- As an example, suppose John takes up still photography as a hobby.
- His resource-limited PC can barely handle the requirements for a photo editing software, say X-photo editor.
- In order to augment his PC's regular performance, he uninstalls the X-photo editor software and purchases a cloud service, which lets him access a virtual machine (VM).
- In his VM, he installs the X-photo editor software, by which he can edit photos efficiently and most importantly, without worrying about burdening his PC or running out of processing resources.
- After six months, John's interest in his hobby grows and he moves on to video editing too.
- For editing his captured videos, he installs video editing software, Y-video editor, in his VM and can edit videos efficiently.

- Additionally, he has the option of installing and using a variety of software for different purposes.

(b) Availability:

- Virtualization creates a logical separation of the resources of multiple entities without any intervention of end users.
- The concept of virtualization makes available a considerable amount of resources as per user requirements.
- The end users feel that there are unlimited resources present dedicatedly for him/her.
- Let us suppose that Jane uses a particular email service.
- Her account has been active for over ten years now; however, it offers limited storage of 2 GB.
- Due to the ever accumulating file attachments in different emails, her 2 GB complimentary space is exhausted.
- However, there is a provision that if she pays \$100 annually, she can attach additional space to her mail service.
- This upgrade allows her to have more storage at her disposal for a considerable time in the future.

(c) Portability:

- Portability signifies the availability of cloud computing services from anywhere in the world, at any instant of time.
- For example, a person flying from the US to the UK still has access to their documents, although they cannot physically access the devices on which the data is stored.
- This has been made possible by platforms such as Google Drive.

d) Elasticity:

- Through the concept of virtualization, an end user can scale-up or scale-down resource utilization as per requirements.
- The end user needs to pay the amount based on their usage.
- For example, Jack rents two VMs in a cloud computing infrastructure from a CSP.
- VM1 has the Ubuntu operating system (OS), on which Jack is simulating a network scenario using Network Simulator-2 (NS2).
- VM2 has Windows 10 OS, on which he is running a MATLAB simulation.
- However, after a few days, Jack feels that his VM2 has served its purpose and is no longer required.
- Consequently, he releases VM2 and, after that, he is only billed for VM1.
- Thus, Jack can scale-up or scale-down his resources in cloud computing, which employs the concept of virtualization.

(ii) Advantages for CSP

➤ Resource Utilization:

- CSP in a cloud computing architecture procures resources on their own or get them from third parties.
- These resources are distributed among different users dynamically as per their requirements.
- A segment of a particular resource provided to a user at a time instant, can be provided to another user at a different time instant.
- In the cloud computing architecture, resources can be re-utilized for multiple users.

➤ Effective Revenue Generation:

- A CSP generates revenue from the end users based on resource utilization.
- As an example, today, a user **A** is utilizing storage facility from a particular CSP.

- The user will release the storage after a few days when his/her requirement is complete.
- The CSP earns some revenue from user A for the utilization of the allocated storage facility.
- In the future, the CSP can provide the same storage facility to a different user, B.
- The CSP can generate revenue from user B for his/her storage utilization.

10.2.2 Types of virtualization

Based on the requirements of the users, virtualization can be categorized as shown in Figure 10.3.

(i) Hardware Virtualization:

- This type of virtualization indicates the sharing of hardware resources among multiple users.
- For example, a single processor appears as many different processors in a cloud computing architecture.
- Different operating systems can be installed in these processors and each of them can work as stand-alone machines.

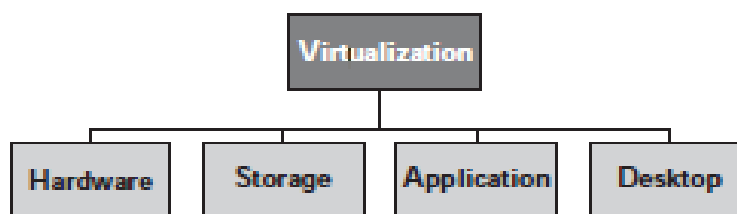


Figure 10.3 Types of virtualization

(ii) Storage Virtualization:

- In storage virtualization, the storage space from different entities are accumulated virtually, and seem like a single storage location.

- Through storage virtualization, a user's documents or files exist in different locations in a distributed fashion.
- However, the users are under the impression that they have a single dedicated storage space provided to them.

(iii) Application Virtualization:

- A single application is stored at the cloud end.
- However, as per requirement, a user can use the application in his/her local computer without ever actually installing the application.
- Similar to storage virtualization, in application virtualization, the users get the impression that applications are stored and executed in their local computer.

(iv) Desktop Virtualization:

- This type of virtualization allows a user to access and utilize the services of a desktop that resides at the cloud.
- The users can use the desktop from their local desktop.

10.3 Cloud Models

- As per the National Institute of Standards and Technology (NIST) and Cloud Computing Standards Road map Working Group, the cloud model can be divided into two parts:
 - (1) Service model and
 - (2) Deployment model
- Further the service model is categorized as:
 - Software-as-a-Service (SaaS),
 - Platform-as-a-Service(PaaS),

- Infrastructure-as-a-Service (IaaS).
- On the other hand, the deployment model is further categorized as:
- Private cloud,
- Community cloud,
- Public cloud,
- Hybrid cloud.

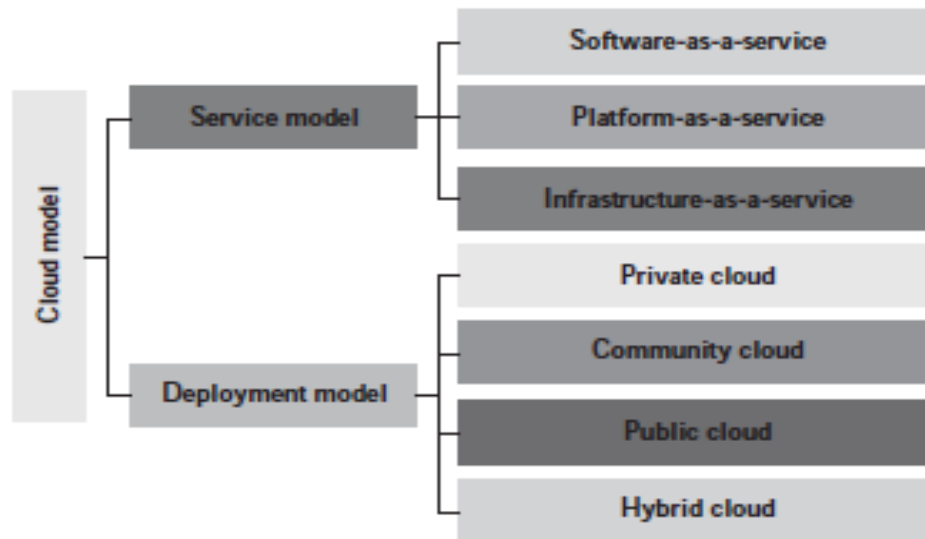


Figure 10.4 Cloud model

(i) Service Model

The service model is depicted in Figure 10.5.

- **Software-as-a-Service (SaaS):**
- This service provides access to different software applications to an end user through Internet connectivity.
- For accessing the service, a user does not need to purchase and install the software applications on his/her local desktop.
- The software is located in a cloud server, from where the services are provided to multiple end users.
- SaaS offers scalability, by which users have the provision to use multiple software applications as per their requirements.

- Additionally, a user does not need to worry about the update of the software applications.
- These software are accessible from any location.
- One example of SaaS is Microsoft Office 365.

- **Platform-as-a-Service (PaaS):**

- PaaS provides a computing platform, by which a user can develop and run different applications.
- The cloud user need not go through the burden of installing and managing the infrastructure such as operating system, storage, and networks.
- However, the users can develop and manage the applications that are running on top of it.
- An example of PaaS is Google App Engine.

- **Infrastructure-as-a-Service (IaaS):**

- IaaS provides infrastructure such as storage, networks, and computing resources.
- A user uses the infrastructure without purchasing the software and other network components.
- In the infrastructure provided by a CSP, a user can use any composition of the operating system and software.
- An example of IaaS is Google Compute Engine.

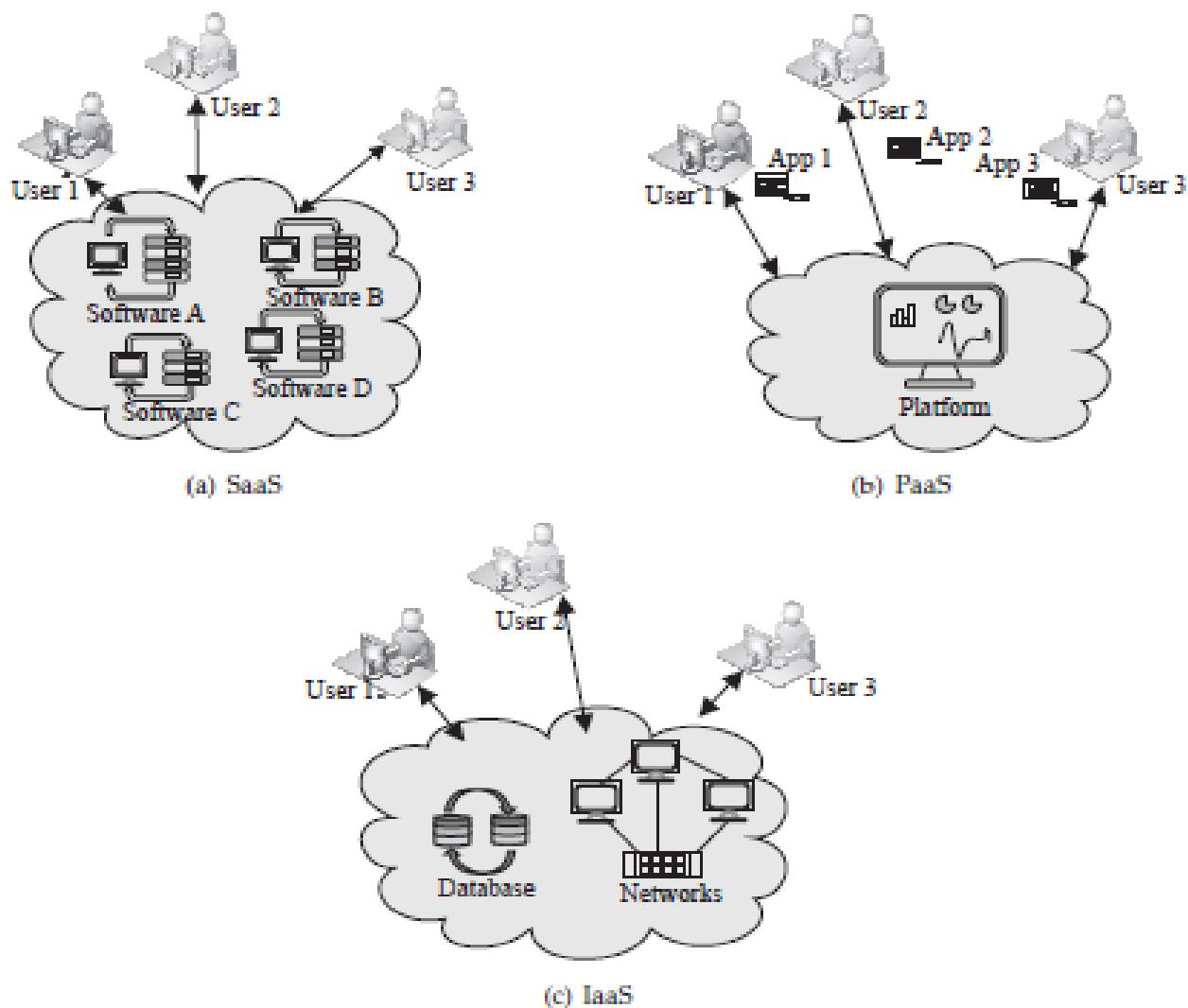


Figure 10.5 Service models

(ii) Deployment Model

Private Cloud:

- This type of cloud is owned explicitly by an end user organization. The internal resources of the organization maintain the private cloud.

Community Cloud:

- This cloud forms with the collaboration of a set of organizations for a specific community.
- For a community cloud, each organization has some shared interests.

Public Cloud:

- The public cloud is owned by a third party organization, which provides services to the common public.
- The service of this cloud is available for any user, on a payment basis.

Hybrid Cloud:

- This type of cloud comprises two or more clouds (private, public, or community).

10.4 Service-Level Agreement in Cloud Computing

- The most important actors in cloud computing are the end user/customer and CSP.
- Cloud computing architecture aims to provide optimal and efficient services to the end users and generate revenue from them as per their usage.
- Therefore, for a clear understanding between CSP and the customer about the services, an agreement is required to be made, which is known as service-level agreement (SLA).
- An SLA provides a detailed description of the services that will be received by the customer.
- Based on the SLA, a customer can be aware of each and every term and condition of the services before availing them.
- An SLA may include multiple organizations for making the legal contract with the customers.

10.4.1 Importance of SLA

- An SLA is essential in cloud computing architecture for both CSP and customers.
- It is important because of the following reasons:

Customer Point of View:

- Each CSP has its SLA, which contains a detailed description of the services.
- If a customer wants to use a cloud service, he/she can compare the SLAs of different organizations.
- Therefore, a customer can choose a preferred CSP based on the SLAs.

CSP Point of View:

- In many cases, certain performance issues may occur for a particular service, because of which a CSP may not be able to provide the services efficiently.
- In such a situation, a CSP can explicitly mention in the SLA that they are not responsible for inefficient service.

10.4.2 Metrics for SLA

- Depending on the type of services, an SLA is constructed with different metrics.
- A few common metrics that are required to be included for constructing an SLA are as follows:

(i) **Availability:** This metric signifies the amount of time the service will be accessible for the customer.

(ii) **Response Time:** The maximum time that will be taken for responding to a customer request is measured by response time.

(iii) **Portability:** This metric indicates the flexibility of transferring the data to another service.

(iv) **Problem Reporting:** How to report a problem, whom and how to be contacted, is explained in this metric.

(v) **Penalty:** The penalty for not meeting the promises mentioned in the SLA.

10.5 Cloud Implementation

10.5.1 Cloud simulation

- With the rapid deployment of IoT infrastructure for different applications, the requirement for cloud computing is also increasing.
- It is challenging to estimate the performance of an IoT system with the cloud before real implementation.
- Real deployment of the cloud is a complex and costly procedure.
- There is a requirement for simulating the system through a cloud simulator before real implementation.
- Many cloud simulators provide pre-deployment test services for repeatable performance evaluation of a system.
- A cloud simulator provides the following advantages to a customer:
 - Pre-deployment test before real implementation,
 - System testing at no cost,
 - Repeatable evaluation of the system,
 - Pre-detection of issues that may affect the system performance,
 - Flexibility to control the environment.
- Currently, different types of cloud simulators are available.
- A few cloud simulators are listed here:

(i) Cloud Sim

(a) Description:

- Cloud Sim is a popular cloud simulator that was developed at the University of Melbourne.
- This simulator is written in a Java-based environment.
- In Cloud Sim, a user is allowed to add or remove resources dynamically during the simulation and evaluate the performance of the scenario.
- **Features:**
- Cloud Sim has different features, which are listed as follows:

- (1) The Cloud Sim simulator provides various cloud computing data centers along with different data center network topologies in a simulation environment.
- (2) Using CloudSim, virtualization of server hosts can be done in a simulation.
- (3) A user is able to allocate virtual machines (VMs) dynamically.
- (4) It allows users to define their own policies for the allocation of host resources to VMs.
- (5) It provides flexibility to add or remove simulation components dynamically.
- (6) A user can stop and resume the simulation at any instant of time.

(ii) Cloud Analyst

(a) Description:

- Cloud Analyst is based on Cloud Sim.
- This simulator provides a graphical user interface (GUI) for simulating a cloud environment, easily.
- The Cloud Analyst is used for simulating large-scale cloud applications.

(b) Features:

- (1) The Cloud Analyst simulator is easy to use due to the presence of the GUI.
- (2) It allows a user to add components and provides a flexible and high level of configuration.
- (3) A user can perform repeated experiments, considering different parameter values.
- (4) It can provide a graphical output, including a chart and table.

(iii) GreenCloud

(a) Description:

- Green Cloud is developed as an extension of a packet level network simulator, NS2.

- This simulator can monitor the energy consumption of different network components such as servers and switches.

(b) Features:

- (1) Green Cloud is an open-source simulator with user-friendly GUI.
- (2) It provides the facility for monitoring the energy consumption of the network and its various components.
- (3) It supports the simulations of cloud network components.
- (4) It enables improved power management schemes.
- (5) It allows a user to manage and configure devices, dynamically, in simulation.

10.5.2 An open-source cloud: Open Stack

- For the real implementation of cloud, there are various open-source cloud platforms available such as Open Stack, Cloud Stack, and Eucalyptus.
- The Open Stack is free software, which provides a cloud IaaS to users.
- A user can easily use this cloud with the help of a GUI-based web interface or through the command line.
- Open Stack supports a vastly scalable cloud system, in which different pre-configured software suites are available.
- The service components of Open Stack along with their functions are depicted in Table 10.1.

Features of Open Stack

- (i) Open Stack allows a user to create and deploy virtual machines.
- (ii) It provides the flexibility of setting up a cloud management environment.
- (iii) Open Stack supports an easy horizontal scaling: dynamic addition or removal of instances for providing services to multiple numbers of users.
- (iv) This cloud platform allows the users to access the source code and share their code to the community.

Table 10.1 Components in OpenStack

Component	Function
Nova	Compute
Neutron	Networking
Cinder	Block storage
Keystone	Identity
Glance	Image
Swift	Object storage
Horizon	Dashboard
Trove	Database
Sahara	Elasticmap reduce
Manila	Shared file system
Designate	DNS
Searchlight	Search
Barbican	Key manager

10.5.3 A commercial cloud: Amazon web services (AWS)

- Besides the open-source cloud, there are various commercial cloud infrastructures available in the market.
- Few of the popular commercial cloud infrastructures are Amazon Web Services (AWS), Microsoft Azure, and Google App Engine.
- A user can launch and manage server instances in AWS.
- A web interface is used to handle the instances.
- Additionally, AWS provides different APIs (application programming interfaces), tools, and utilities for users.
- Amazon AWS follows the pay-per-use model.
- This cloud infrastructure provides a virtual computing environment, where different configurations, such as CPU, memory, storage, and networking capacity are available.

Features of AWS

- (i) It provides flexibility to scale and manage the server capacity.

- (ii) AWS provides control to OS and deployment software.
 - (iii) It follows the pay-per-use model.
 - (iv) The cloud allows a user to establish connectivity between the physical network and private virtual network.
 - (v) The developer tools in this cloud infrastructure help a user for fast development and deployment of the software.
 - (vi) AWS provides excellent management tools, which help a user to monitor and automate different components of the cloud.
 - (vii) The cloud provides machine learning facilities, which are very useful for data scientists and developers.
 - (viii) For extracting meaning from data, analytics play an important role.
- AWS also provides a data analytics platform.

10.6 Sensor -Cloud: Sensors-as-a-Service

- In a sensor-cloud, virtualization of sensors plays an essential role in providing services to multiple users.
- In a sensor-cloud architecture, multiple users receive services from different a sensor nodes, simultaneously.
- The users remain oblivious to the fact that a set of sensor nodes is not dedicated solely to them for their application requirements.
- In reality, a particular sensor may be used for serving multiple user applications, simultaneously.
- The main aim of sensor-cloud infrastructure is to provide an opportunity for the common mass to use Wireless Sensor Networks (WSNs) on a payment basis.
- Sensor-cloud architecture follows the pay-per-use model.

10.6.1 Importance of sensor-cloud

- The sensor-cloud infrastructure is based on the concept of cloud computing, in which a user application is served by a set of homogeneous or heterogeneous sensor nodes.

- These sensor nodes are selected from a common pool of sensor nodes, as per the requirement of user applications.
- Using the sensor-cloud infrastructure, a user receives data for an application from multiple sensor nodes without owning them.
- Figures 10.6 depict the usage of sensor nodes using traditional WSN and sensor cloud infrastructure.
- The advantages of sensor-cloud over traditional WSN.

Case Study:

- John is a farmer, and he has a significantly vast farmable area with him.
- As manual supervision of the entire field is very difficult, he has planned to deploy a WSN in his farming field.
- Before purchasing the WSN, he has to decide which sensors should be used in his fields for sensing the different agricultural parameters.
- Additionally, he has to decide the type and number of other components such as an electronics circuit board and communication module required along with the sensors.
- As there are numerous vendors, it is challenging for him to choose the correct (in terms of quality and cost) vendor, as well as the sensor owner from whom the WSN will be procured.
- He finally decides the type of sensors along with the other components that are required for monitoring his agricultural field.
- Now, John faces the difficulty of optimally planning the sensor node deployment in his fields.
- After going through these hurdles, he decides on the number of sensor nodes that are required for monitoring his field.
- Finally, John procures the WSNs from a vendor.
- After procurement, he deploys the sensor nodes and connects different components.

- As WSN consists of different electronic components, he has to maintain the WSN after its deployment.
- After three months, as his requirement of agricultural field monitoring is completed, he removes the WSN from the agricultural field.
- Six months later, John plans to use the WSN that was deployed in the agricultural field for home surveillance.
- As the agriculture application is different from the home surveillance application, the sensor required for the system also changes.
- Thus, John has to go through all the steps again, including maintenance, deployment, and hardware management, for the surveillance system. Thus, we observe that the users face different responsibilities for using a WSN for an application.
- In such a situation, if sensor-cloud architecture is present, John can easily use WSNs for his application on a rental basis.
- Moreover, with sensor-cloud, John can easily switch the application without any manual intervention.
- On the other end, service providers of the sensor-cloud infrastructure may serve multiple users with the same sensors and earn profit.

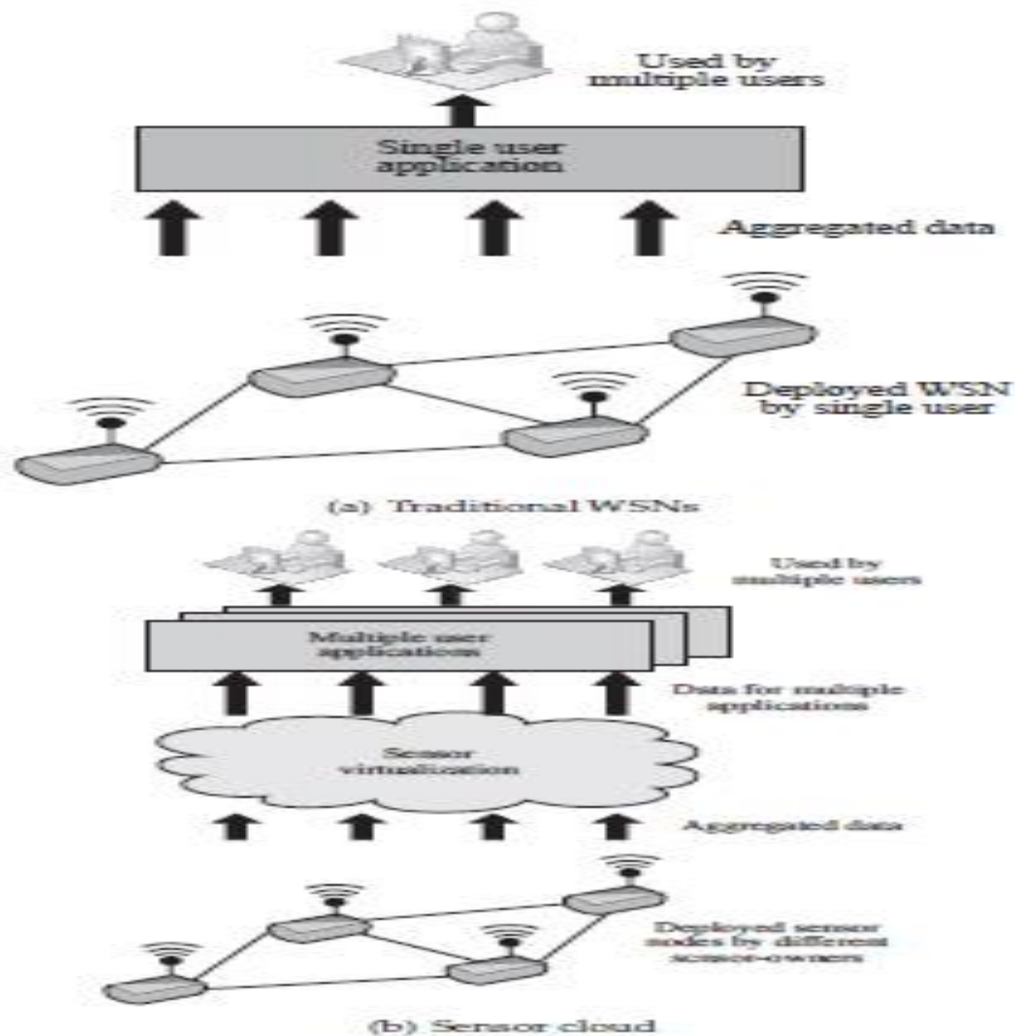


Figure 10.6 Traditional WSN versus sensor-cloud

10.6.2 Architecture of a sensor-cloud platform

- In a traditional cloud computing architecture, two actors, cloud service provider (CSP) and end users (customer) play the key role.
- Unlike cloud computing, in sensor-cloud architecture, the sensor owners play an important role along with the service provider and end users.
- However, a service provider in sensor-cloud architecture is known as a sensor-cloud service provider (SCSP).
- The detailed architecture of a sensor-cloud is depicted in Figure 10.7.

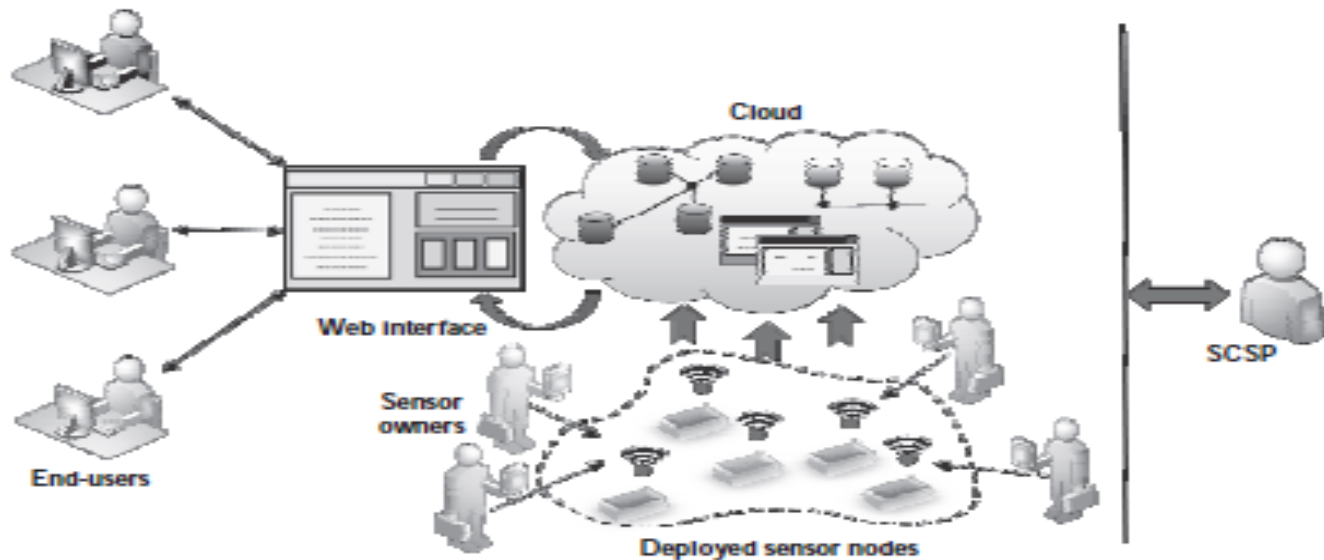


Figure 10.7 Architecture of a sensor-cloud platform

Actors in sensor-cloud architecture

- Typically, in sensor-cloud architecture, three actors are present.
- We briefly describe the role of each actor.

(i) End User:

- This actor is also known as a customer of the sensor-cloud services.
- Typically, an end user registers him/herself with the infrastructure through a Web portal.
- Thereafter, he/she chooses the template of the services that are available in the sensor-cloud architecture to which he/she is registered.
- Finally, through the Web portal, the end user receives the services, as shown in Figure 10.7.
- Based on the type and usage duration of service, the end user pays the charges to the SCSP.

(ii) Sensor Owner:

- The deployment of the sensors is essential in order to provide services to the end users.
- These sensors in a sensor cloud architecture are owned and deployed by the sensor owners, as depicted in Figure 10.7.
- A particular sensor owner can own multiple homogeneous or heterogeneous sensor nodes.
- Based on the requirements of the users, these sensor nodes are virtualized and assigned to serving multiple applications at the same time.
- On the other hand, a sensor owner receives rent depending upon the duration and usage of his/her sensor node(s).

(iii) Sensor-Cloud Service Provider (SCSP):

- An SCSP is responsible for managing the entire sensor-cloud infrastructure (including management of sensor owners and end users handling, resource handling, database management, cloud handling etc.), centrally.
- The SCSP receives rent from end users with the help of a pre-defined pricing model.
- The pricing scheme may include the infrastructure cost, sensor owners' rent, and the revenue of the SCSP.
- Typically, different algorithms are used for managing the entire infrastructure.
- The SCSP receives the rent from the end users and shares a partial amount with the sensor owners.
- The remaining amount is used for maintaining the infrastructure.
- In the process, the SCSP earns a certain amount of revenue from the payment of the end users.
- The sensor-cloud architecture of two view points:
 - (i) User organizational view
 - (ii) Real architectural view.

Different views of sensor-cloud architecture are shown in Figure 10.8.

(i) User Organizational View:

- This view of sensor-cloud architecture is simple.
- In a sensor-cloud, end users interact with a Web interface for selecting templates of the services.
- Thereafter, the end users through the Web interface receive the services.
- In this architecture, an end user is unaware of the complex processes that are running at the back end.

(ii) Real Architectural View:

- The complex processing of sensor-cloud architecture is visualized through this view.
- The processes include sensor allocation, data extraction from the sensors, virtualization of sensor nodes, maintenance of the infrastructure, data center management, data caching, and others.
- For each process, there is a specific algorithm or scheme.

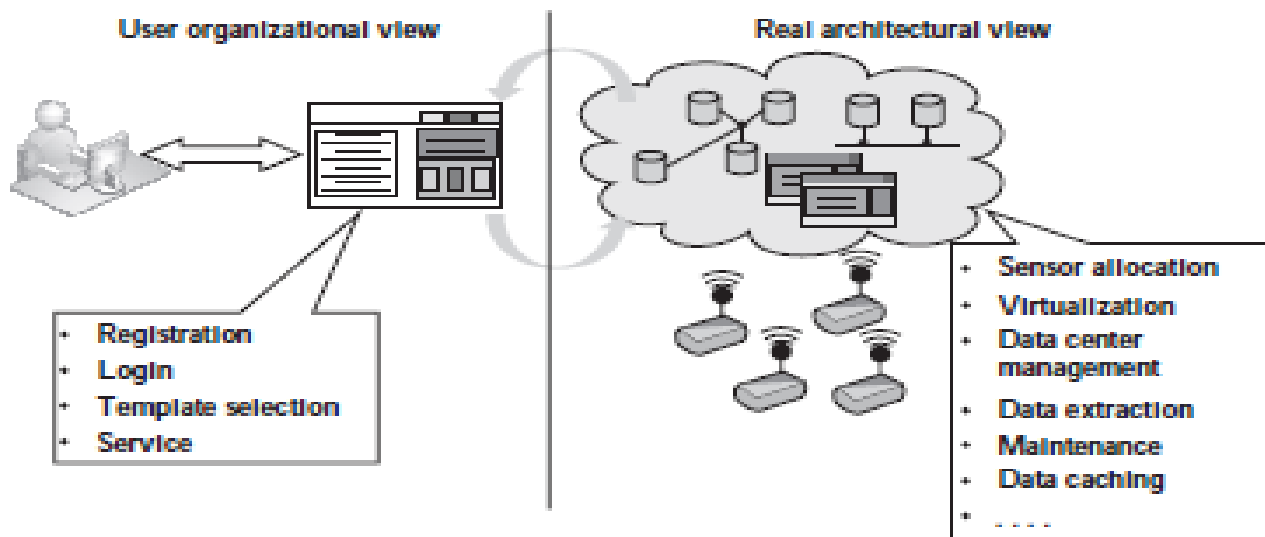


Figure 10.8 Sensor-cloud architecture